M. Caner Tol

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Education

Ph.D. in Electrical and Computer Engineering — Aug 2019 - Dec 2024
Worcester Polytechnic Institute, Worcester, MA
Advisor: Prof. Berk Sunar

M.S. in Electrical and Computer Engineering — Aug 2019 - May 2022
Worcester Polytechnic Institute, Worcester, MA

B.S. in Electrical and Electronics Engineering — Sep 2014 - Jun 2019
Middle East Technical University, Ankara, TR

Experience

Research Associate — Worcester, MA
Worcester Polytechnic Institute — Feb 2025 - Current

- Open Tools, Interfaces and Metrics for Implementation Security Testing
- optimist-ose.org

Hardware Security Intern — Santa Clara, CA
NVIDIA — May 2024 - Aug 2024

- Evaluated EM side-channel attacks on GPUs for ML model extraction and defenses against them.

Hardware Security Intern — Santa Clara, CA
NVIDIA — May 2022 - Aug 2022

- Developed a ML-based tool for automatic detection of side-channel leakage on CUDA libraries using GPU hardware performance counters.

Research and Teaching Assistant — Worcester, MA
Vernam Lab, Worcester Polytechnic Institute — Aug 2019 - Dec 2024

- Research on Microarchitectural side-channel, fault injection attacks, and the applications of Transformer models, generative AI, and reinforcement learning on discovery, detection, and mitigation of these attacks.

Skills

HW Security Microarchitectural side-channels, Transient Execution Attacks, Rowhammer, Voltage glitching, EM side-channels, TVLA, Oscilloscope

Programming: ARM/x86 Assembly, C/C++, Python, CUDA, Bash, MATLAB, Git

Binary Analysis: Linux Perf, PAPI, Intel Pin, objdump, Ghidra

Libraries: PyTorch, TensorFlow, Keras, Scipy

Research

M. C. Tol, K. Derya, and B. Sunar, "μRL: Discovering Transient Execution Vulnerabilities Using Reinforcement Learning", 2025 *[Under Review].*

K. Derya, **M. C. Tol**, and B. Sunar, "Fault+Probe: A Generic Rowhammer-based Bit Recovery Attack", in *Proceedings of the 2025 ACM Asia CCS, Hanoi, Vietnam, 2025.*

A. J. Adiletta, **M. C. Tol**, and B. Sunar, "LeapFrog: The Rowhammer Instruction Skip Attack", *hardwear.io USA 2024, IEEE European Symposium on Security and Privacy Venice, Italy 2025.*

M. C. Tol, and B. Sunar, "ZeroLeak: Automated Side-Channel Patching in Source Code Using LLMs", *Real World Crypto, Toronto, Canada, 2024.* and *ESORICS, Bydgoszcz, Poland, 2024.*

A. J. Adiletta*, **M. C. Tol***, Y. Doroz, and B. Sunar, "Mayhem: Targeted Corruption of Register and Stack Variables", in *Proceedings of the 2024 ACM Asia CCS, Singapore, 2024.* 3rd place in poster competition @NEHWS23.

M. C. Tol, S. Islam, A. J. Adiletta, B. Sunar, and Z. Zhang, "Don't Knock! Rowhammer at the Backdoor of DNN Models", in *Proceedings of the 2023 IEEE/IFIP International Conference on Dependable Systems and Network, Porto, Portugal, 2023.* 1st place in poster competition @NEHWS22.

K. Mus, Y. Doroz, **M. C. Tol**, K. Rahman, and B. Sunar, "Jolt: Recovering TLS Signing Keys via Faults", in *Proceedings of the 2023 IEEE Symposium on Security and Privacy, San Francisco, CA, 2023.*

M. C. Tol, B. Gulmezoglu, K. Yurtseven, and B. Sunar, "FastSpec: Scalable Generation and Detection of Spectre Gadgets Using Neural Embeddings", in *Proceedings of the 2021 IEEE European Symposium on Security and Privacy, Vienna, Austria, 2021.*

L. Amorós, S. M. Hafiz, K. Lee, and **M. C. Tol**, "Gimme That Model!: A Trusted ML Model Trading Protocol", *Protecting Privacy through Homomorphic Encryption, 2021.*

B. Gulmezoglu, A. Zankl, **M. C. Tol**, S. Islam, T. Eisenbarth, and B. Sunar, "Undermining User Privacy on Mobile Devices Using AI", in *Proceedings of the 2019 ACM Asia CCS, Auckland, New Zealand, 2019.*

Awards

Best Poster Award — Apr 2022
New England Hardware Security Day

IoT Application Award — Jun 2019
The Senior Engineering Design Course Committee, METU EE

Honor Student — Jun 2019
Middle East Technical University

Services

Reviewer at **IEEE Transactions on Computers** — January 2025

Artifact Evaluation Committee Member at **ACM CCS** — May 2024

Reviewer at **IEEE Transactions on Information Forensics and Security** — December 2024

Reviewer at **IEEE Transactions on Emerging Topics in Computing** — October 2023

Reviewer at **The Computer Journal, Oxford University Press** — May 2023